ACID - BASIC Character

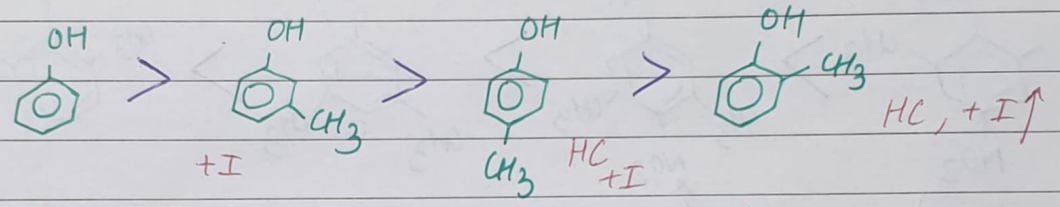
Acidic char ⇒ stability of C.B → eWG/eDG
→ eN of atom
→ Period ⇒ eN
Group ⇒ size

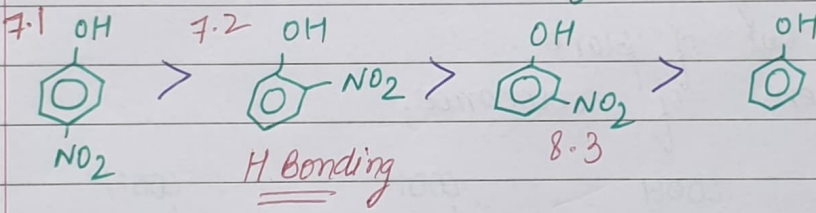
Exceptional cases :

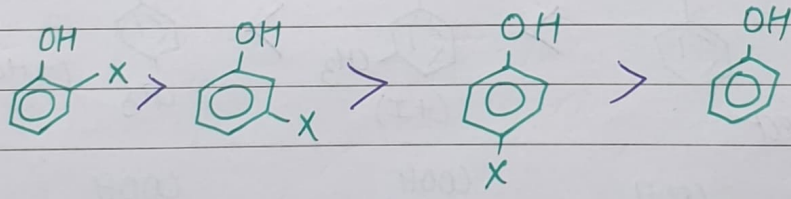
1. $HC\equiv CH$ > Ph_3CH > Ph_2CH_2 > $PhCH_3$ > $CH_2=CHCH_3$ > CH_3CH_3
pKa = 25 (33) (50)

2. CH_3-CH_2-COOH < $CH_2=CH-COOH$ < $CH\equiv C-COOH$
15.5 15.74

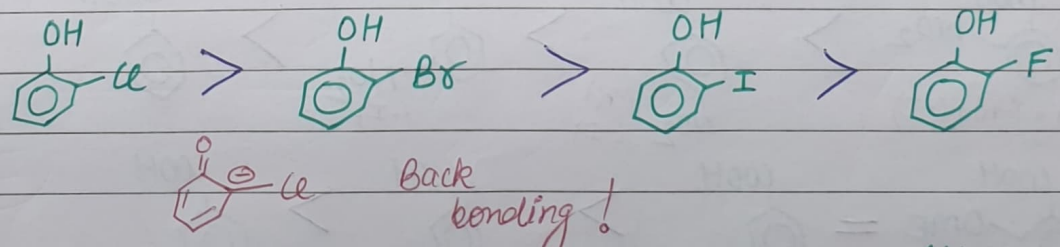
3. CH_3OH > H_2O > CH_3CH_2OH 16

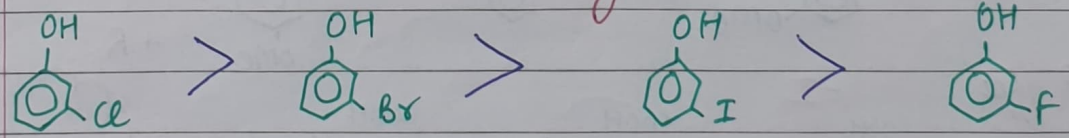
4. 

5. 

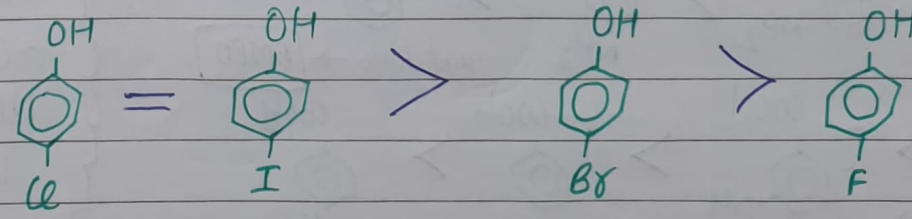
6. 

Simple (-I)

7. 

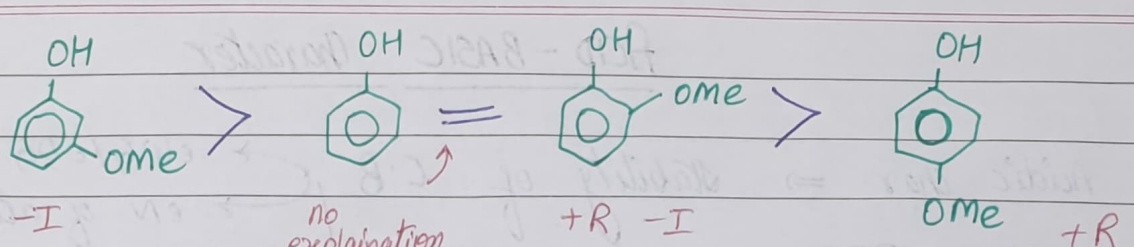
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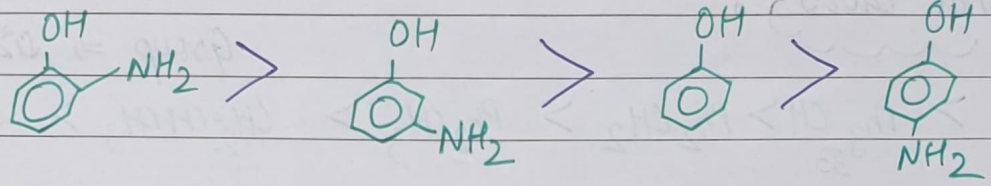
no explaination ∴

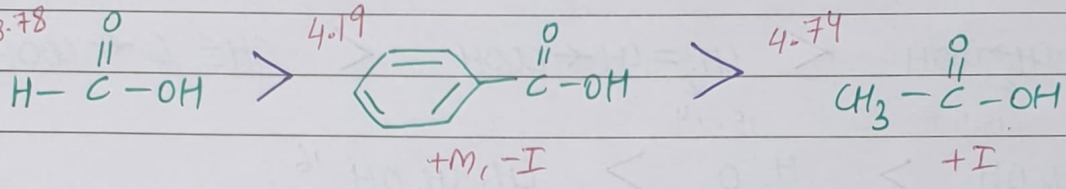
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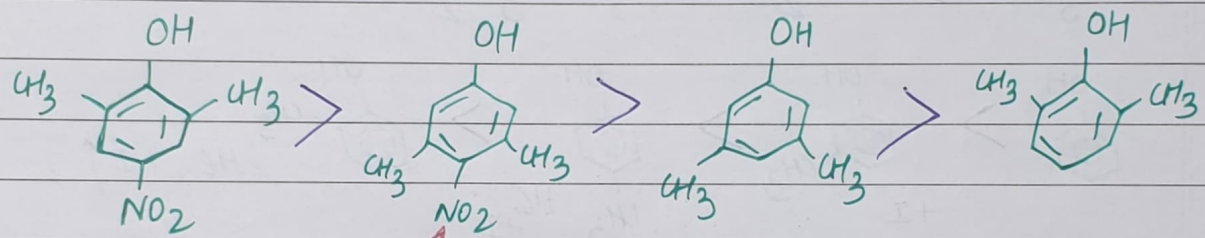
no explaination ∴

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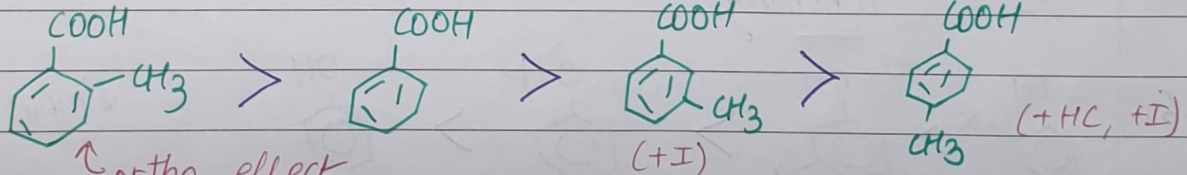
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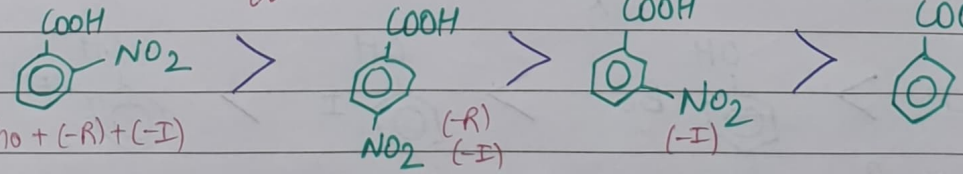
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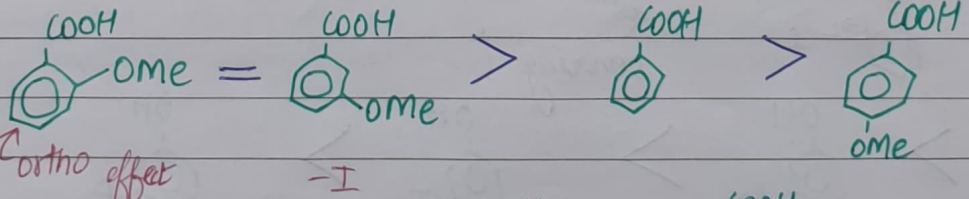
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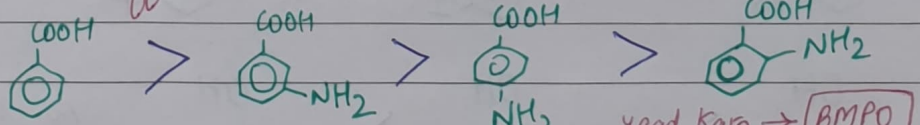
13. 

SIR steric inhibition goes out of plane of resonance.

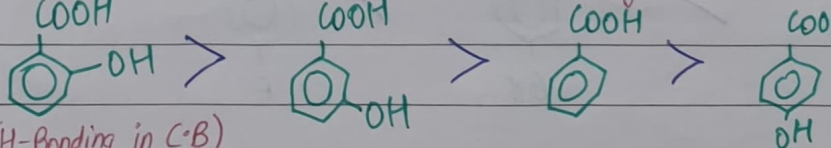
14. 

15. 

16. 

*17. 

yaad karo → BMPO

*18. 

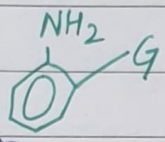
-OH, -NH2 -CN DO NOT show Ortho effect

Basic char:

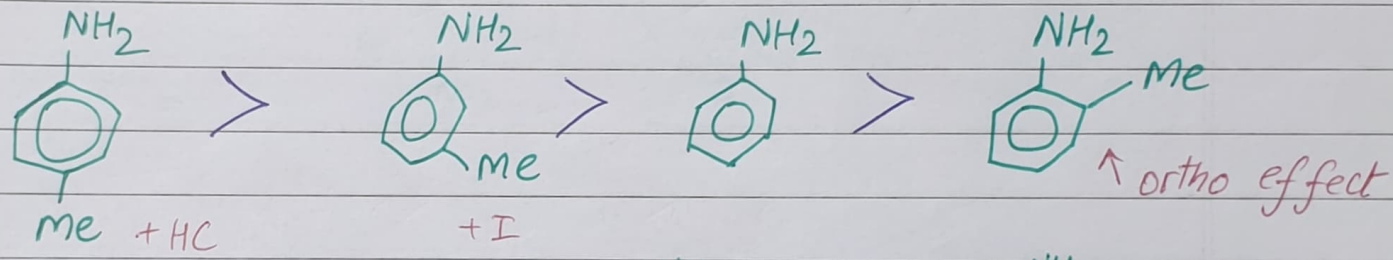
gas / non-polar $\Rightarrow$ $R_3\ddot{N}$ (3°) > $R_2\ddot{N}H$ (2°) > $R\ddot{N}H_2$ (1°) > $\ddot{N}H_3$

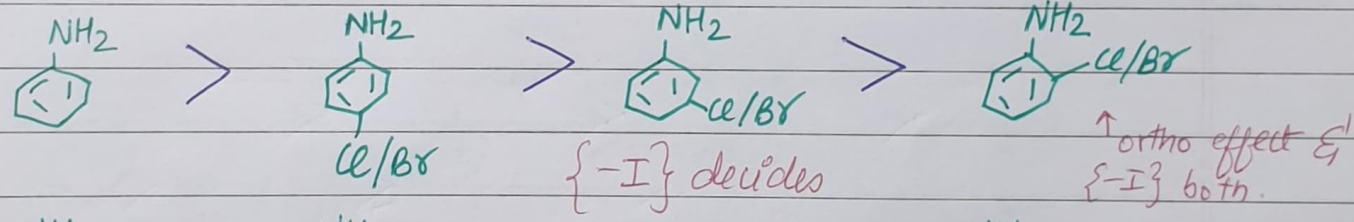
Polar $\Rightarrow$ $R = CH_3 \Rightarrow$ 2° > 1° > 3° > NH_3

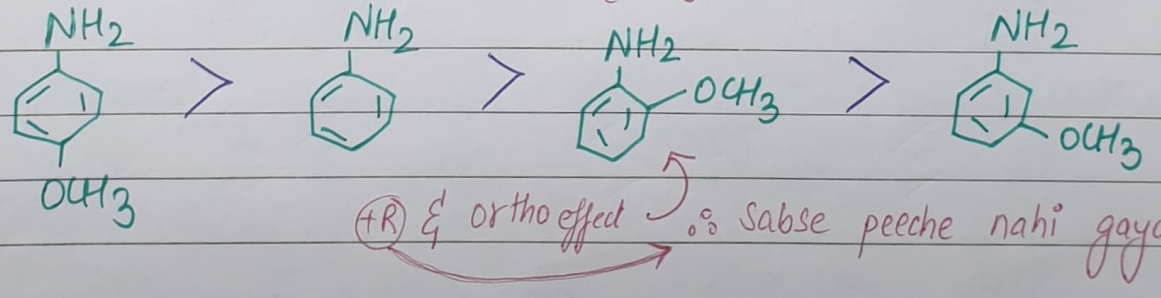
$R > CH_3 \Rightarrow$ 2° > 3° > 1° > NH_3

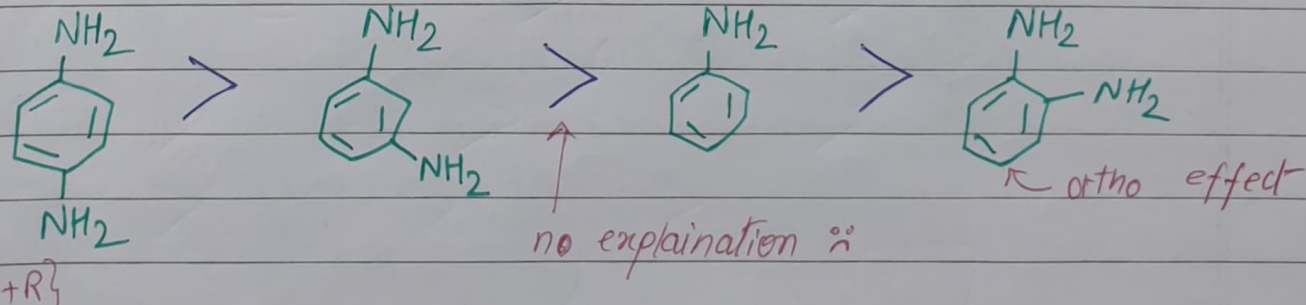
* Ortho effect $\Rightarrow$  decreases basicity due to S.H.

{Exceptional cases:}

1. 

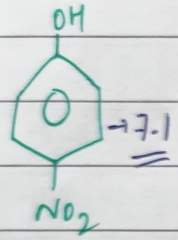
2. 

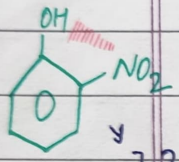
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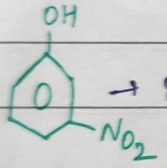
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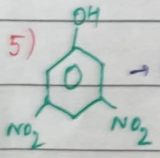
pKa

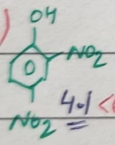
1)  → 9.9

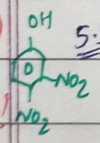
2)  → 7.1

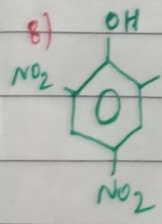
3)  → 7.2

4)  → 8.3

5)  → 6.7 > 6.3!

6)  4.1 < 6.3!

7)  5.3 < 6.3!

8)  ≃ 0.2

picric acid

1) H_2O → 15.74

2) H_2CO_3 → I) 6.35 (HO-C(=O)-OH) II) 10.3

3) H_3O^+ → −1.74

4) NH_3 → 37

5) CH_4 → 50

* CH_3-CH_3 → 50

$CH_2=CH_2$ → 45

$HC\equiv CH$ → 25

RSH → 10−11

HF → 3.2

HCl → −7

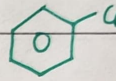
HBr → −9

HI → −10

CARBOXYLIC ACID

* $R-C(=O)-OH$ → 4-6

CH_3COOH → 4.74

 → 4.19

$HCOOH$ → 3.78

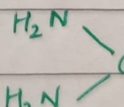
ALCOHOLS

* ROH → 16-18

CH_3CH_2OH → 16

 CH_3OH → 15.4

CH_3OH → 15.5

 $pK_b \approx 0$

Guanidine

DATE
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★ $CH_3-\overset{O}{\overset{\|}{C}}-H$ 13.5

$Me-\overset{O}{\overset{\|}{C}}-Me$ 20

$Me-\overset{O}{\overset{\|}{C}}-O-Et$ 25

★ $Ph-\underset{\underset{Ph}{|}}{\overset{Ph}{\overset{|}{CH}}}$ 33